



Web Programming II Project

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EVENT MANAGEMENT & TICKET BOOKING SYSTEM

INTRODUCTION

The Event Management and Ticket Booking System is a web-based application designed to simplify the process of creating, managing, and booking events. It allows users to perform CRUD operations (Create, Read, Update, Delete) on events and also includes a ticket booking feature.

The system is built using Node.js, Express.js, EJS templating engine, and PostgreSQL as the database.

OBJECTIVES

The main objectives of this project are:

- To allow users to create and manage events easily
- To store event data in a structured database
- To enable ticket booking functionality
- To practice full-stack web development using modern technologies
- To demonstrate backend routing and database integration

TECHNOLOGIES USED

Frontend:

- EJS (Embedded JavaScript Templates)
- HTML5
- Bootstrap 5 (for styling)

Backend:

- Node.js
- Express.js

Database:

- PostgreSQL

Tools:

- VS Code
- pgAdmin
- Git & GitHub

SYSTEM FEATURES**Create Event**

Users can create an event by entering:

- Event name
- Event date
- Location

View Events

All events are displayed in descending order.

Edit Event

Users can update event details.

Delete Event

Events can be removed from the system.

Ticket Booking

Users can book tickets, reducing available ticket count.

DATABASE DESIGN

Database Name:

eventdb

Table: events

```
CREATE TABLE events (  
  id SERIAL PRIMARY KEY,  
  name VARCHAR(255) NOT NULL,  
  location VARCHAR(255) NOT NULL,  
  date DATE NOT NULL,  
  tickets INT DEFAULT 50  
);
```

Table Description:

Column Type		Description
id	SERIAL	Primary Key
name	VARCHAR(255)	Event name
location	VARCHAR(255)	Event location
date	DATE	Event date
tickets	INT	Available tickets

SYSTEM ARCHITECTURE

User (Browser)



EJS Views (Frontend UI)



Express Routes (Backend Logic)



PostgreSQL Database

PROJECT STRUCTURE

Event management and ticket booking/

```
|
|├─ app.js
|├─ server.js
|├─ db.js
|
|├─ routes/
|  └─ eventRoutes.js
|
|├─ views/
|  ├── events.ejs
|  ├── create.ejs
|  ├── edit.ejs
|  └─ eventDetails.ejs
|
|├─ public/
|
└─ package.json
```

KEY CODE EXPLANATION

Server Setup (server.js)

- Starts the Express server
- Runs on port 3000

Middleware (app.js)

- Parses JSON and form data
- Sets view engine to EJS
- Loads routes

Database Connection (db.js)

- Connects Node.js to PostgreSQL using pg module

Event Routes (eventRoutes.js)

Handles:

GET /events

- Fetches all events from database

GET /events/new

- Opens create event page

POST /events

- Inserts event into database

GET /events/edit/:id

- Loads edit page

POST /events/edit/:id

- Updates event

POST /events/delete/:id

- Deletes event

POST /events/book/:id

- Reduces ticket count

MAIN ISSUE FIXED IN PROJECT

During development, the following issue was solved:

- **Error:**
relation "events" does not exist
- **Cause:**
The PostgreSQL table was not created.
- **Solution:**
Created the `events` table manually in eventdb database.

HOW TO RUN PROJECT

- `npm install`
`node server.js`
- Then open:
- <http://localhost:3000/events>

GIT WORKFLOW

- `git add .`
`git commit -m "Final event management system"`
`git push origin main`

CONCLUSION

- This project successfully demonstrates a full-stack web application with backend routing, database integration, and frontend rendering. It provides a strong foundation for understanding CRUD operations and real-world web application architecture.